

DELAWARE COASTAL, DE, USA

WMO: 724093

Lat: 38.690N

Lon: 75.363W

Elev: 46

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 13764

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.7	19.2	-1.7	5.0	19.6	2.9	6.4	23.0	26.1	50.7	21.8	50.5	5.3	310	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	93.0	75.6	90.7	74.8	88.3	73.8	78.5	87.9	77.4	86.1	76.3	84.3	9.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.0	136.1	82.2	74.9	131.2	81.2	73.6	125.4	80.0	42.0	88.3	40.7	86.2	39.7	84.2	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	18.3	16.4	9.6	97.2	5.4	2.8	5.7	99.2	2.6	100.9	-0.4	102.5	-4.3	104.5
			8.0	80.9	5.1	1.1	4.4	81.7	1.4	82.3	-1.5	82.9	-5.2	83.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.4	36.4	38.7	45.4	55.6	64.6	73.2	77.8	76.4	70.2	59.4	49.0	41.0
	DBStd	16.58	10.50	9.43	9.60	8.78	7.85	6.29	5.16	4.76	6.17	8.19	8.42	9.25
	HDD50	1463	438	335	209	37	1	0	0	0	0	13	118	312
	HDD65	4118	885	736	611	305	108	9	0	0	22	214	482	746
	CDD50	4171	18	19	66	207	454	696	863	819	605	304	89	31
	CDD65	1352	0	0	3	25	96	255	398	354	177	40	3	1
	CDH74	11844	2	1	31	240	858	2298	3888	3060	1205	251	10	0
	CDH80	4322	0	0	4	62	261	856	1639	1119	331	50	0	0
	WSAvg	6.9	7.2	7.5	8.3	8.5	7.4	6.7	6.1	5.6	6.2	6.3	6.3	6.6
	PrecAvg	46.1	3.2	3.0	3.7	3.4	3.6	3.9	4.6	5.0	4.2	3.3	3.2	3.7
(15) Precipitation	PrecMax	61.1	5.1	6.1	7.1	5.6	7.2	10.8	7.4	9.6	8.9	7.6	5.9	7.4
	PrecMin	36.9	1.3	0.8	0.4	1.5	1.0	0.9	1.7	0.6	0.4	0.0	0.4	1.7
	PrecStd	7.2	1.1	1.3	1.7	1.0	1.8	2.6	1.6	2.4	2.3	2.2	1.7	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.3	70.3	78.2	86.2	90.0	94.5	97.4	94.7	91.3	85.0	74.9	70.3
		MCWB	61.4	60.2	62.4	66.1	71.3	75.1	77.7	77.1	74.6	71.4	64.5	64.0
	2%	DB	64.0	64.4	72.0	81.1	86.5	91.0	93.5	91.3	87.3	80.6	71.1	65.7
		MCWB	59.1	55.7	59.6	64.5	70.6	74.2	75.9	76.1	73.0	69.2	62.0	61.3
	5%	DB	59.0	59.2	66.2	75.3	82.5	88.2	91.2	89.1	83.7	76.0	66.6	61.4
		MCWB	53.9	51.9	56.6	62.2	68.9	73.3	75.1	74.8	71.7	67.0	60.4	57.0
	10%	DB	53.5	54.2	61.6	71.3	79.2	85.0	88.5	86.2	80.9	72.6	63.2	56.4
		MCWB	48.4	47.7	53.4	60.1	67.2	71.8	74.6	73.6	70.5	64.9	57.1	52.0
	0.4%	WB	63.6	63.0	64.8	70.1	74.4	78.0	80.4	80.1	78.2	75.1	67.5	65.6
		MCDB	66.9	68.0	74.6	80.0	84.7	89.1	91.1	89.2	85.4	80.6	71.5	68.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	59.9	57.9	61.9	67.2	72.4	76.4	78.8	78.4	76.2	72.3	64.5	61.8
		MCDB	63.8	62.7	68.8	77.3	82.7	86.5	89.0	87.0	82.0	77.9	68.5	65.0
	5%	WB	54.7	53.7	58.6	64.5	70.8	75.3	77.6	77.2	74.8	69.7	61.7	57.7
		MCDB	58.5	57.7	65.5	72.6	80.2	84.4	87.0	85.1	80.0	74.3	65.8	60.8
	10%	WB	48.5	48.0	54.5	62.1	68.9	74.2	76.5	76.0	73.3	66.6	58.2	52.9
		MCDB	52.3	53.5	60.2	69.5	77.0	82.2	85.0	83.1	78.1	71.0	62.3	55.5
(35) Mean Daily Temperature Range	5% DB	MDBR	16.0	17.4	18.3	20.1	19.1	18.0	17.6	17.1	17.5	18.6	18.6	16.7
		MCDBR	22.4	24.1	25.3	25.7	23.3	21.2	20.4	19.5	20.6	21.1	21.7	21.1
		MCWBR	18.4	18.0	15.4	13.3	10.7	8.4	7.1	7.4	8.9	11.3	14.4	18.4
	5% WB	MCDBR	21.5	22.0	22.8	22.0	21.0	18.3	17.7	16.7	15.5	17.3	18.2	19.9
		MCWBR	19.2	19.2	16.5	13.4	10.3	8.1	7.0	7.2	7.9	10.9	14.5	18.9
(40) Clear-Sky Solar Irradiance	taub		0.316	0.320	0.358	0.394	0.443	0.488	0.508	0.503	0.412	0.360	0.327	0.315
	taud		2.479	2.468	2.399	2.318	2.212	2.134	2.095	2.094	2.325	2.472	2.516	2.522
	Ebn at Noon		273	288	286	280	267	255	248	246	264	269	267	266
	Edh at Noon		25	29	35	40	45	49	50	49	37	29	24	23
(44) All-Sky Solar Radiation	RadAvg		675	943	1263	1615	1825	1995	1948	1738	1421	1062	764	593
	RadStd		59	98	105	118	203	153	94	112	144	108	69	57

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.19	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (69 neighbors)		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+250	+148

Nomenclature: See separate page